

Customer Shopping Behavior Analysis

Data Analysis Project Report
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1. Project Overview

This project explores customer shopping behavior using transactional data. The objective was to identify spending patterns, customer segments, product preferences, and subscription trends to help businesses make data-driven decisions.

Tools Used: Python (Pandas), SQL (PostgreSQL), Power BI, Microsoft Excel

2. Dataset Summary

Total Records: 3,900 transactions

Total Columns: 18 features

Missing Data: 37 missing values in Review Rating column

Key features in the dataset:

- Customer Demographics: Age, Gender, Location, Subscription Status
- Purchase Details: Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
- Shopping Behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Data Cleaning & Preparation (Python)

The raw dataset was loaded and processed in Python using the Pandas library. The following steps were performed:

- Data Loading: Imported the CSV dataset using `pandas read_csv()`
- Initial Exploration: Used `df.info()` and `df.describe()` to understand the data structure and summary statistics
- Missing Value Treatment: Imputed 37 missing values in the Review Rating column using the median rating of each product category
- Column Standardization: Renamed all columns to snake_case format for consistent querying
- Feature Engineering: Created age_group column by binning customer ages into Young Adult, Adult, Middle-aged, and Senior groups
- Feature Engineering: Created purchase_frequency_days column from the purchase frequency data
- Data Consistency Check: Found that discount_applied and promo_code_used were redundant — dropped promo_code_used
- Database Integration: Loaded the cleaned DataFrame into PostgreSQL using SQLAlchemy for SQL-based analysis

4. Business Analysis using SQL (PostgreSQL)

After loading clean data into PostgreSQL, the following 10 business questions were answered using SQL queries:

Query 1: Revenue by Gender

Compared total revenue generated by male vs. female customers.

Gender	Total Revenue (USD)
Male	\$1,57,890

Female	\$75,191
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Query 2: High-Spending Discount Users

Identified 839 customers who used discounts but still spent above the average purchase amount — showing that discount users are not always low-value customers.

Query 3: Top 5 Products by Average Rating

Product	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Query 4: Shipping Type Comparison

Shipping Type	Avg Purchase Amount (USD)
Express	\$60.48
Standard	\$58.46

Query 5: Subscribers vs Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
Subscribed	1,053	\$59.49	\$62,645
Not Subscribed	2,847	\$59.87	\$1,70,436

Query 6: Top 5 Discount-Dependent Products

Product	Discount Rate (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Query 7: Customer Segmentation

Segment	Number of Customers
Loyal	3,116

Returning	701
New	83

Query 8: Top 3 Products per Category

Identified the most purchased products within each category using SQL RANK() window function.

- Accessories: Jewelry (171), Sunglasses (161), Belt (161)
- Clothing: Blouse (171), Pants (171), Shirt (169)
- Footwear: Sandals (160), Shoes (150), Sneakers (145)
- Outerwear: Jacket (163), Coat (161)

Query 9: Repeat Buyers & Subscription

Checked whether customers with more than 5 purchases are more likely to subscribe. Result: 2,518 repeat buyers were non-subscribers vs. 958 subscribers — suggesting a loyalty program opportunity.

Query 10: Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

5. Power BI Dashboard

An interactive dashboard was built in Power BI to present all key insights visually with filters for Category, Gender, Subscription Status, and Shipping Type.

Key KPIs displayed:

- Total Customers: 3,900
- Average Purchase Amount: \$59.76
- Average Review Rating: 3.75

Visuals included:

- Donut chart: Subscription status breakdown (Yes 27%, No 73%)
- Bar charts: Revenue by Category and Revenue by Age Group
- Comparative charts: Sales by Category and Sales by Age Group



6. Business Recommendations

- **Boost Subscriptions:** Only 27% of customers are subscribed. Exclusive benefits and loyalty rewards can increase this number significantly.
- **Loyalty Programs:** With 3,116 loyal customers, a structured rewards program can retain and upsell to this group.
- **Discount Policy Review:** Products like Hat and Sneakers have ~50% discount rates — pricing strategy needs review to protect margins.
- **Product Promotion:** Promote top-rated products (Gloves, Sandals, Boots) more prominently in marketing campaigns.
- **Target Young Adults:** This age group contributes the highest revenue (\$62,143) — marketing should focus here.
- **Express Shipping Promotion:** Express shipping users spend slightly more (\$60.48) — promoting it could increase average order value.